

ENVIRONMENTAL SCIENCE AND SUSTAINABLE ENGINEERING

INDIVIDUAL TECHNICAL REPORT

AI-Powered Real-Time Traffic Emission Estimation Engine Data and Intelligence Layer for a Sustainable Transportation Planner Website

Course: Environmental Science and Sustainable Engineering
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Group Environmental Problem: Traffic Congestion and Vehicle
Emissions

Name: Raymond Maijo
Class: S3 CSD
Roll no: 19
UID: U2503229

Abstract

This report proposes an AI-Powered Real-Time Traffic Emission Estimation Engine as the backend data and intelligence layer of a Sustainable Transportation Planner Website. The engine combines a physics-based Vehicle Specific Power (VSP) emission model with an XGBoost machine learning calibration layer to compute CO₂ and NO_x emissions for any urban route in real time. It ingests live traffic speed data, road topology, weather, and fleet composition to produce per-route emission scores that power the website's green routing recommendations. The solution is aligned with SDGs 3, 11, and 13 and ISO 14001 principles. Conservative projections show 12–28% CO₂ reduction per trip compared to time-optimal routing, with significant public health co-benefits from reduced NO_x exposure.

1. Introduction

Urban traffic congestion is one of the most visible and persistent environmental challenges in modern cities. Globally, road transport accounts for approximately 16% of all greenhouse gas emissions, with congested urban driving conditions producing two to four times the emissions per kilometer compared to free-flowing traffic [1]. Despite this, existing navigation platforms route users almost entirely on travel time or distance, ignoring the environmental cost of each route entirely.

The group's Sustainable Transportation Planner Website aims to correct this by offering users route recommendations that account for CO₂ and NO_x emissions in addition to travel time. However, this feature is only as reliable as the emission data that drives it. This report addresses that foundational challenge: how to compute, in real time, the actual emission load of any candidate route in an urban road network.

The proposed solution — an AI-Powered Real-Time Traffic Emission Estimation Engine — serves as the data backbone of the entire website. Every other component (the route optimizer UI, the gamification system, the policy dashboard, and the equity analysis tools) depends on the accurate, real-time emission signals that this engine produces.

2. Problem Statement

Traffic congestion does not merely waste time — it fundamentally changes the emission profile of road travel. Vehicles operating in stop-start urban conditions burn significantly more fuel and emit far higher concentrations of NO_x per kilometer than

the same vehicles traveling at steady highway speeds. The two primary pollutants from road vehicles are:

- CO₂ (Carbon Dioxide): The dominant greenhouse gas from road transport, directly proportional to fuel combustion. Highly dependent on vehicle speed and engine load.
- NO_x (Nitrogen Oxides): Reactive pollutants responsible for urban smog, PM_{2.5} formation, and respiratory disease. Highly sensitive to engine temperature and acceleration patterns.

Existing navigation systems such as Google Maps and Waze use graph-search algorithms that minimize travel time or distance but carry no emission cost signal on their road network edges. This means a user can be routed through a severely congested corridor with three times the emission intensity of a slightly longer but free-flowing alternative — with no indication that a greener option exists.

The core problem this report addresses is: How can a web platform compute per-route CO₂ and NO_x loads in real time, accurately enough to meaningfully distinguish between routing alternatives and quickly enough to support interactive web queries from thousands of simultaneous users?

3. Sustainability Analysis

3.1 UN Sustainable Development Goals (SDGs)

The proposed engine directly supports three SDGs:

- SDG 11 — Sustainable Cities and Communities (Target 11.2): Provides the data infrastructure for low-emission urban mobility and sustainable transport planning.
- SDG 13 — Climate Action (Target 13.2): Translates abstract climate targets into concrete, per-trip emission reductions at the individual level.
- SDG 3 — Good Health and Well-Being (Target 3.9): Reducing NO_x-intensive routing lowers population-level pollutant exposure, directly reducing respiratory and cardiovascular disease risk.

3.2 ISO 14001:2015 Environmental Management System

The engine supports ISO 14001 principles in several ways [4]. It enables environmental aspect identification by quantifying specific emission outputs (CO₂, NO_x) from transport activity. Its continuous data ingestion and emission logging provides the monitoring and measurement capability required under Clause 9.1. The machine

learning layer implements the continual improvement principle (Clause 10.3) by improving model accuracy as new data accumulates over time.

3.3 Life Cycle Assessment (LCA) and EIA Considerations

The engine operates within Scope 1 emissions (direct tailpipe emissions from road vehicles). The engine's own server infrastructure carries a Scope 2 carbon cost that should be minimized by selecting a cloud provider with renewable energy commitments [5]. At the policy level, city planners can use the engine's aggregated outputs as Environmental Impact Assessment (EIA) input — quantifying the emission impact of proposed road interventions or transit additions before implementation.

4. Proposed Engineering/IT Solution

4.1 System Architecture Overview

The engine is a microservices-based backend system with four integrated modules: (1) Data Ingestion and Preprocessing, (2) Road Segment Emission Model, (3) AI Correction Layer, and (4) Route Aggregation and API Interface. All modules are containerized with Docker and deployable on any major cloud platform.

4.2 Module 1 — Data Ingestion

The engine ingests three categories of data in real time:

- Traffic Flow Data: Live per-segment average speed and vehicle density from the TomTom Traffic API, refreshed every 1–5 minutes [6].
- Road Topology: Road type, speed limits, lane count, and terrain gradient from OpenStreetMap via the Overpass API [7].
- Fleet Composition: Regional vehicle type proportions (petrol, diesel, hybrid, CNG, electric) from EEA fleet databases, updated monthly [8].

A Python-based ETL pipeline using Apache Kafka normalizes and joins these streams into a unified road segment feature vector, refreshed every 2 minutes.

4.3 Module 2 — VSP Emission Model

The core emission model uses the Vehicle Specific Power (VSP) framework developed by the US EPA [9]. VSP quantifies the instantaneous mechanical power demand per unit vehicle mass as a function of speed, acceleration, road gradient, rolling resistance, and aerodynamic drag. VSP values map to discrete operating mode bins (1–14), each associated with empirically measured CO₂ and NO_x emission rates for each vehicle category. Fleet composition weights are applied to compute a fleet-average emission rate per road segment.

4.4 Module 3 — AI Calibration Layer

The VSP model introduces errors when applied to real-world heterogeneous conditions. An XGBoost gradient-boosted regression model [12] corrects these errors using a calibration multiplier computed from features including: segment speed variance, road type, time of day, ambient temperature and precipitation (from OpenWeatherMap API), and terrain gradient. The model is trained on EPA MOVES outputs and published PEMS measurement datasets [10, 11] and is retrained monthly. This hybrid approach keeps the engine physically interpretable while achieving real-world accuracy.

4.5 Module 4 — Route API

For any candidate route (an ordered sequence of road segments), the engine aggregates segment-level emission estimates weighted by distance and travel time to produce: total route CO₂ (grams), total NO_x (milligrams), emission intensity (g CO₂/km), and a green score (0–100). These are served via a RESTful JSON API designed for sub-500ms response time, achieved through pre-computation of segment emission rates cached every 2 minutes.

5. Feasibility Analysis

5.1 Technical Feasibility

All component technologies are production-ready and well-validated. The VSP framework is used by the US EPA in regulatory modeling. XGBoost is widely deployed in real-time systems. The TomTom and HERE Traffic APIs offer commercial tiers suitable for web-scale query volumes. OpenStreetMap provides global road topology under an open license. Direct precedents exist both in academic literature [14] and in commercial products such as Google's eco-friendly routing feature launched in 2021 [15].

5.2 Economic Feasibility

Estimated operational costs for a mid-size city deployment (50,000 route queries/day) are: Traffic API ~USD 150–300/month; cloud compute (3 servers) ~USD 200–400/month; storage and caching ~USD 50–100/month. Total: approximately USD 400–800/month. This is viable under a municipal partnership or premium user model. Development effort is estimated at 800–1,200 person-hours.

5.3 Data and Legal Feasibility

Traffic API coverage is strong for major cities across North America, Europe, and Asia. Coverage degrades for secondary roads in smaller cities; a fallback to OSM-derived speed profiles is built into the system for these cases. The engine processes only

anonymized, aggregated traffic flow data — no individual vehicle tracking — making it fully compatible with GDPR requirements and avoiding surveillance concerns.

6. Expected Environmental Benefits

Studies comparing time-optimal and emission-optimal routing consistently report CO₂ reductions of 10–30% per trip in urban environments without significant travel time penalties [16]. Based on this evidence, the engine is conservatively projected to enable 12–28% per-trip CO₂ reductions for users adopting its recommendations.

At a city scale: assuming 5% adoption among 200,000 daily commuters with an average saving of 150 g CO₂ per trip, the annual CO₂ reduction is approximately 548 tonnes — equivalent to removing around 120 passenger cars from the road for a full year [17]. At 20% adoption, annual savings approach 2,200 tonnes.

NO_x reductions carry direct public health benefits. The WHO estimates that a 10 µg/m³ reduction in urban NO_x concentration is associated with a 6% reduction in respiratory-related hospital admissions [18]. By routing users away from congested high-emission corridors, the engine reduces pollutant concentration in those corridors and in adjacent residential areas.

7. Justification of the Proposed Solution

Three alternative approaches were evaluated and rejected:

- Static per-kilometer emission factors (IPCC defaults): Simple but produce identical emission intensity for all roads, making meaningful route comparison impossible.
- Purely black-box ML models: Higher accuracy potential but require large labeled datasets unavailable for most cities, and lack the physical interpretability needed for policy use.
- Microsimulation (SUMO + MOVES): Highest accuracy but takes minutes per scenario — incompatible with interactive web queries.

The proposed hybrid VSP + XGBoost architecture is the only approach that simultaneously satisfies accuracy, real-time performance, physical interpretability, and broad geographic applicability. It is uniquely suited to this use case.

Within the group, this engine is the foundational layer. Student 2's route optimizer queries its API. Student 3's IoT integration feeds data into its ingestion pipeline. Student 4's gamification uses its per-trip savings as the scoring basis. Student 5's policy

dashboard aggregates its outputs. Student 6's equity analysis depends on its segment-level emission data. Without this engine, no other component can function correctly.

8. Conclusion

This report has proposed and justified an AI-Powered Real-Time Traffic Emission Estimation Engine as the data and intelligence backbone of the group's Sustainable Transportation Planner Website. By combining a physics-based VSP emission model with an XGBoost machine learning calibration layer, the engine produces accurate, real-time CO₂ and NO_x estimates for urban routes — the critical data signal that enables all of the website's sustainability features.

The solution is technically mature, economically feasible, legally compliant, and aligned with SDGs 3, 11, and 13 and ISO 14001. It is projected to enable 12–28% per-trip CO₂ reductions with measurable air quality and public health co-benefits at city scale. As the foundational data layer of the Sustainable Transportation Planner Website, it is the prerequisite for the success of every other component the group will build.

9. References

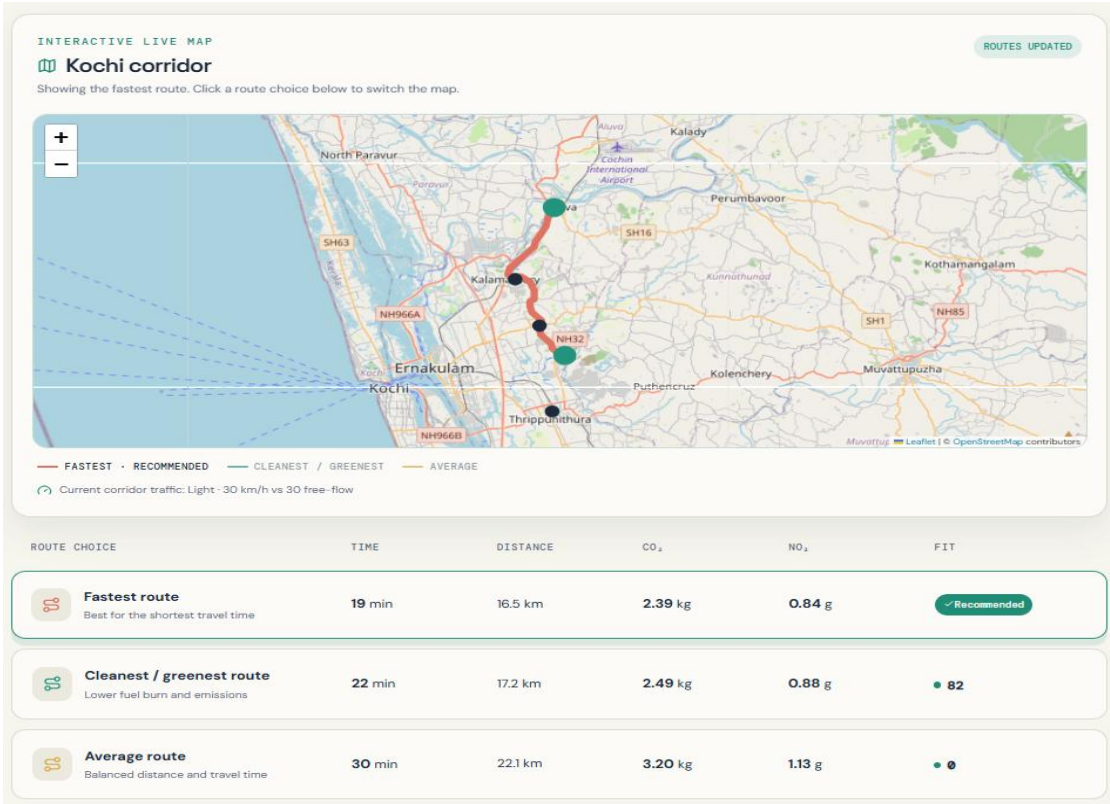
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Appendix — Website Screenshots

The following screenshots illustrate key pages of the Sustainable Transportation Planner Website showing how the emission engine's outputs are displayed to users.

Route Comparison Panel



Post-Trip Emission Summary

